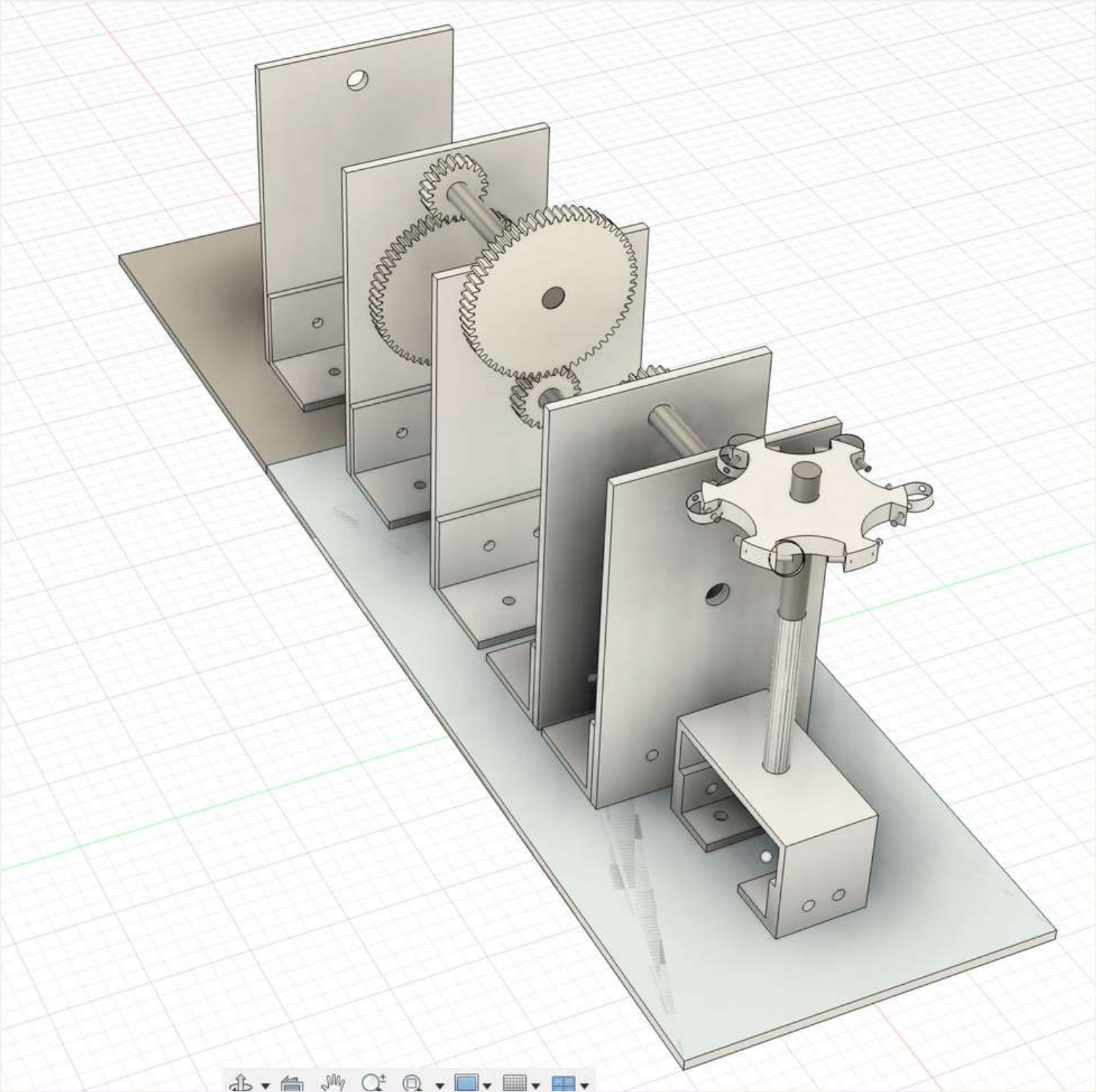


CLINICAL CENTRIFUGE



MEMBERS

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ROLL NO

240116
240667
241196
240235
240875
221207
241040

Motivation & Problem Statement

Context:

- Rural India often lacks reliable electricity and medical infrastructure.
- Clinical centrifuges are essential for diagnostic tests (blood, urine, etc.).

Problem:

- Most centrifuges rely on electricity.
- High cost and maintenance limit their use in rural labs.

Goal:

- Design and fabricate a manual centrifuge that is low-cost, robust, and independent of power supply.

Working Model



**Hand crank
feed**



Spur gears assembly

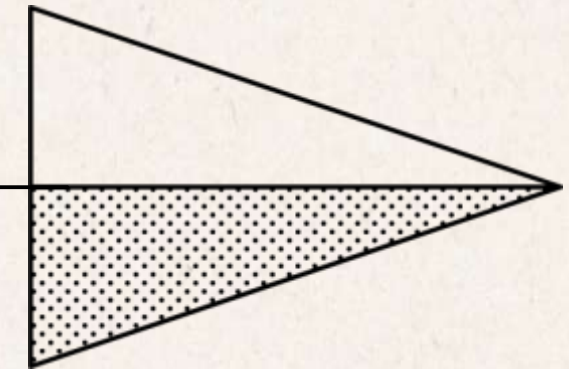
4 stages, 1:3 ratio each
→ $\sim 81\times$ RPM increase



**Bevel gears to
change axis**



**Rotor spins
test tubes**



Concept & Design Overview

Design process:

- Visualizing and creating the CAD model → Creating all the parts individually → Assembly of all the individual parts → Alignment of parts and tuning → Medical calibration and testing.

Key Features:

- Mechanical energy only, no motor.
- Compact and reliable transmission system.
- Modular design for easy maintenance.

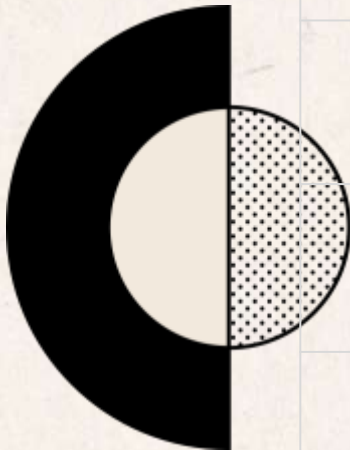
Challenges we faced during the project

- Misalignment of axles
- jammed gears
- Vibration at High RPM
- Clearance issue with the test tubes and the L plate
- Friction between the components

How we tackled the issues

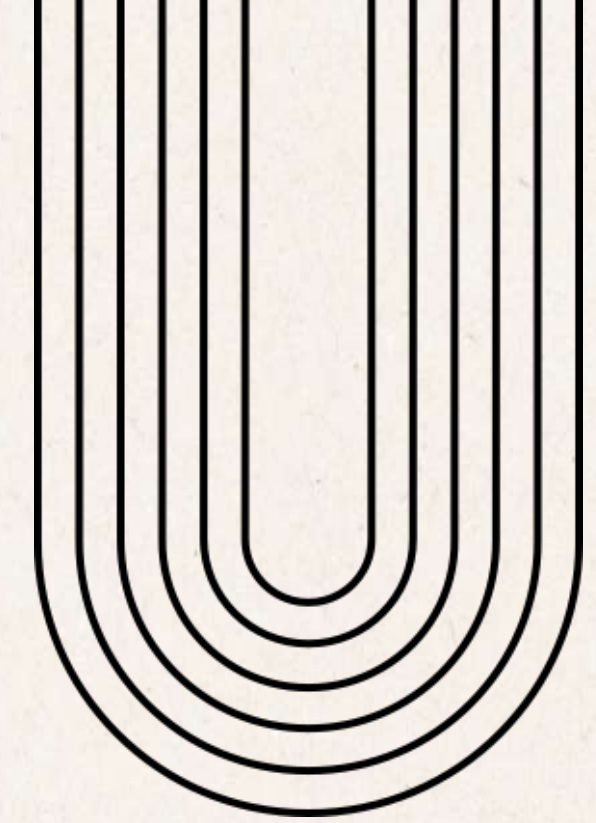
- We raised the L plates using washers for better alignment.
- Tightening axles tolerances solved the jamming issue.
- extra supports were welded at critical places.
- Rotor was raised above the rest of the assembly.
- Adding more lubrication did the job.

Components & Assembly



Component	Quantity	Description
Base Plate	1	Foundation for setup
L-Support / L-Plate	6 each	Frame structure
Spur Gears (20 T & 60 T)	4 each	Gear ratio = 1:3
Bevel Gears (20 T)	2	Axis change
Handle	1	Manual drive
Main Rotor	1	Holds test tubes
Test Tube Holders	6	3D-printed mounts
Axles & Bearings	6	transmit motion

Manufacturing Processes



Operations Involved:

- Cutting and drilling of plates.
- Turning and facing for shafts.
- Milling for gear alignment slots.
- Gear manufacturing (1.5 Module spur & bevel gears).
- 3D printing of test-tube holders and rotor mounts.
- Manual assembly, alignment, and testing.

Results & Advantages

Performance:

- Achieved $\sim 81\times$ RPM amplification via 4 spur gear pairs.
- Smooth transmission and stable rotor rotation.

Advantages:

- Runs without electricity.
- Portable, low-cost, and robust.
- Can be adapted for motor drive if power available.

Conclusions & Future Improvements

Learnings:

- Integrated design → fabrication workflow.
- Practical understanding of machining, tolerances, and gear design.

Future Improvements:

- Incorporate speed governor or tachometer.
- Use lightweight materials for ease of use.

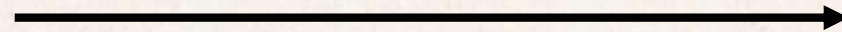
Acknowledgment:

- Thanks to Rahul sir, TA212 Lab Staff, Instructors, and Workshop Technicians.

HERE IS A WORKING MODEL

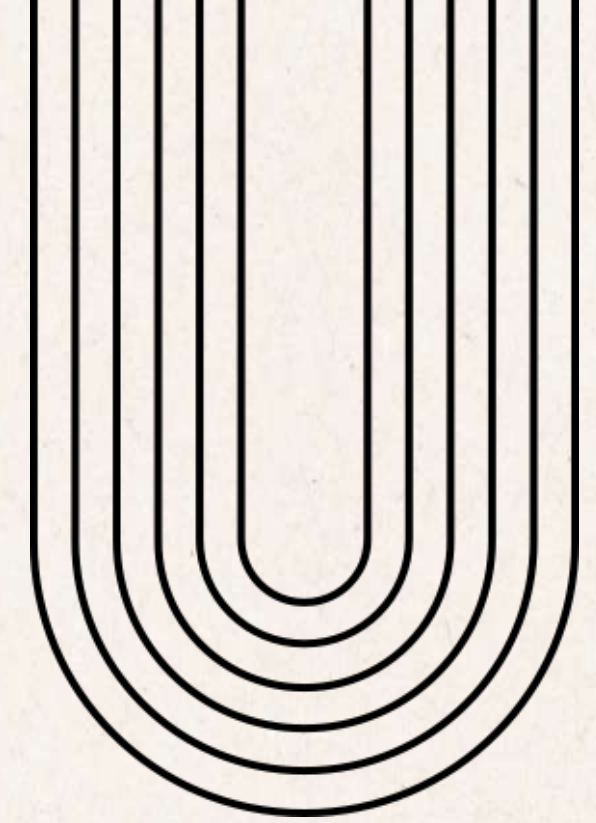


BEFORE CENTRIFUGATION



AFTER CENTRIFUGATION

Learnings from our project



- We learnt how to use appropriate tools for the right job.
- We learnt how to conduct lathing, drilling, milling, turning, facing and other ways to form a certain object.
- We learnt how to test, analyse and implement solutions to problems in our initial design.
- And most importantly we learnt how to work effectively in a team.

Thank you
